

LEE'S SUMMIT, MO, USA

WMO: 720306

Lat: 38.960N

Lon: 94.371W

Elev: 997

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53879

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	9.3	-5.5	4.3	7.8	-1.8	5.2	11.0	27.0	35.7	24.9	34.4	8.5	320	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	95.5	75.3	92.5	75.2	89.9	74.7	78.6	89.9	77.4	88.4	76.2	86.5	10.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	137.9	86.0	74.1	132.1	84.5	72.9	126.7	83.1	42.9	89.8	41.6	88.3	40.3	86.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.0	20.5	18.5	-1.4	99.6	4.7	3.6	-4.8	102.1	-7.5	104.2	-10.1	106.2	-13.5	108.8		
			-2.4	80.9	4.5	1.2	-5.6	81.8	-8.2	82.5	-10.7	83.1	-14.0	84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	31.2	33.8	46.4	55.8	65.8	75.5	78.9	77.0	70.1	57.4	45.7	34.6
	DBStd	19.27	12.20	12.28	11.64	9.60	8.34	5.71	5.82	5.92	7.37	9.56	10.26	10.75
	HDD50	2037	588	471	207	52	4	0	0	0	0	37	194	484
	HDD65	4729	1047	874	582	296	97	3	0	2	33	274	580	941
	CDD50	4276	6	17	96	227	494	764	897	835	602	265	65	8
	CDD65	1494	0	0	6	21	122	317	432	372	185	37	2	0
	CDH74	14340	0	2	35	168	918	3067	4720	3632	1518	273	7	0
	CDH80	5702	0	0	2	26	221	1180	2143	1552	512	66	0	0
Wind	WSAvg	9.0	10.0	10.1	10.7	10.5	8.6	8.5	7.3	7.0	7.5	8.8	9.8	9.7
Precipitation	PrecAvg	42.4	1.4	1.5	2.7	4.1	5.4	5.7	4.1	4.3	5.1	3.8	2.6	1.7
	PrecMax	60.0	3.6	4.1	9.4	10.2	13.3	10.9	13.2	16.7	16.1	9.3	8.7	4.9
	PrecMin	26.0	0.0	0.0	0.4	0.4	1.4	1.0	0.2	0.0	0.6	0.2	0.0	0.0
	PrecStd	8.7	1.0	1.0	1.7	2.4	2.8	2.7	2.5	3.1	3.6	2.2	1.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.4	71.1	78.1	83.8	88.4	95.5	100.7	98.8	92.9	86.0	73.8	65.6
		MCWB	53.9	55.0	62.7	66.5	71.1	75.9	73.9	74.5	73.3	71.8	60.5	58.3
	2%	DB	59.3	64.4	73.4	78.6	85.1	91.8	96.5	94.5	88.9	81.1	69.9	59.8
		MCWB	49.2	53.1	60.1	63.2	69.9	74.9	75.2	75.0	73.6	67.0	58.3	52.2
	5%	DB	53.9	58.0	69.4	75.1	82.5	89.5	93.2	91.3	85.9	76.3	65.3	54.5
		MCWB	45.1	49.1	57.7	61.3	68.6	74.6	75.8	74.9	72.5	64.7	55.3	47.9
	10%	DB	48.1	52.1	64.6	71.7	79.5	86.9	90.2	88.2	82.4	72.6	61.0	50.2
		MCWB	41.2	44.4	54.6	59.7	67.5	73.4	75.2	74.3	70.3	62.6	52.4	44.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.6	59.3	65.4	68.4	74.7	79.1	80.5	80.0	77.2	73.3	64.5	60.3
		MCDB	61.9	65.7	75.2	81.2	85.4	89.5	92.1	91.8	87.3	82.2	70.5	64.5
	2%	WB	50.7	54.6	62.7	65.9	72.4	77.4	79.1	78.3	75.2	70.7	60.0	53.9
		MCDB	56.9	61.8	70.6	75.4	82.0	88.9	90.5	89.7	85.3	77.7	66.4	57.8
	5%	WB	46.0	49.8	59.8	63.6	70.5	76.0	77.8	76.9	73.8	67.9	57.2	49.0
		MCDB	52.1	57.6	67.0	71.9	79.4	86.4	88.6	87.7	83.0	73.4	63.5	53.5
	10%	WB	41.2	44.4	55.7	61.0	68.9	74.4	76.5	75.4	72.2	64.2	53.6	44.2
		MCDB	48.5	52.1	62.4	69.8	77.2	84.4	86.7	85.1	80.8	70.6	60.0	49.6
Mean Daily Temperature Range	5% DB	MDBR	17.1	17.7	18.6	18.9	17.5	17.5	17.4	17.5	18.3	18.3	17.4	16.0
		MCDBR	24.4	26.1	23.3	22.7	20.3	19.4	20.1	20.9	19.3	21.5	22.6	21.6
		MCWBR	16.4	17.0	13.3	13.1	9.8	8.3	6.8	8.0	8.3	11.0	14.4	15.8
	5% WB	MCDBR	22.4	24.2	19.3	19.6	18.2	18.1	17.5	18.3	17.6	17.9	19.9	20.3
		MCWBR	16.7	17.8	13.1	13.4	9.7	9.0	8.2	8.4	8.5	11.3	15.1	17.5
Clear-Sky Solar Irradiance	taub	0.293	0.305	0.334	0.384	0.425	0.426	0.437	0.437	0.385	0.343	0.302	0.289	
	taud	2.472	2.444	2.417	2.295	2.238	2.269	2.268	2.258	2.266	2.362	2.470	2.530	2.509
	Ebn at Noon	282	293	293	283	272	271	267	264	272	275	277	275	
	Edh at Noon	25	30	34	41	44	43	43	41	35	29	23	23	
All-Sky Solar Radiation	RadAvg	697	940	1224	1562	1801	2043	2037	1801	1519	1079	757	603	
	RadStd	74	103	107	100	160	158	127	97	101	121	75	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page